

MILESTONE 4 REPORT

CliniScan – Chest X-ray

Link: <https://huggingface.co/spaces/vishnudharsh-07/cliniscan-ai>

1. Introduction

CliniScan is an AI-powered system designed to analyze chest X-ray images using deep learning techniques. The system assists in detecting lung diseases by combining classification, object detection, and explainability.

2. Objectives

- Integrate classification and detection models
- Implement Grad-CAM explainability
- Develop Streamlit UI
- Provide real-time analysis
- Generate PDF reports

3. System Architecture

Classification: EfficientNet-B0 (15 classes)

Detection: YOLOv8 (14 classes)

Explainability: Grad-CAM heatmaps

4. Implementation

Backend: predict.py, load_models.py, gradcam.py, report.py

Frontend: Streamlit UI with navigation and dashboard

Workflow: Upload → Process → Display → Download report

5. Dataset

VinBigData Chest X-ray dataset used in YOLO and classification formats.

6. Output

- Diagnosis with confidence
- Detection bounding boxes
- Grad-CAM heatmap

- Clinical interpretation
- PDF report

7. Features

AI classification, YOLO detection, Grad-CAM, premium UI, PDF reports

8. Limitations

Diffuse diseases may not show bounding boxes. Performance depends on dataset and hardware.

9. Deployment

Deployed using Hugging Face Spaces with Docker and Streamlit.

10. Conclusion

Successfully integrated models and UI into a working AI system.

11. Future Work

Improve accuracy, add patient data, deploy with GPU, integrate with hospital systems.